UNIT-II

1. What Is a Data Warehouse(DW)? Discuss key features of DW.
2. Explain the Differences between Operational Database Systems and Data Warehouses.
3. Discuss three-tier data warehousing architecture.
4. Explain the following models: Enterprise Warehouse, Data Mart, and Virtual Warehouse
5. Data cleaning, data transformation, refresh
6. Explain Metadata Repository of DW.
7. What is a Data Cube? Explain Multidimensional Data Cube Model.
8. Discuss Snowflake schema, fact constellation, starnet query model of Multidimensional Data model.
9. Explain the Role of Concept Hierarchies in DW.
10. Explain different aggregate functions used to measure of data in multidimensional way.
11. Explain OLAP operations
12. For a data cube with the three dimensions *time, location, and item,* which category does the function variance belong to? Describe how to compute it if the cube is partitioned into many chunks.
13. What is Materialization? Explain materialized views (using cuboids) in DW.
14. Discuss Indexing in OLAP Data for data accessing in DW.
15. Discuss different OLAP Server Architectures.
16. Explain multidimensional Data Generalization by Attribute-Oriented Induction.
17. What are the differences between the three main types of data warehouse usage: information processing, analytical processing, and data mining? Discuss the OLAP mining (OLAM).
18. In data warehouse technology, a multiple dimensional view can be implemented by a relational database technique (ROLAP), by a multidimensional database technique (MOLAP), or by a hybrid database technique (HOLAP).

    (a) Briefly describe each implementation technique.

    (b) For each technique, explain how each of the following functions may be implemented:

        i. The generation of a data warehouse (including aggregation)

        ii. Roll-up

        iii. Drill-down

        iv. Incremental updating
19. A data cube, C, has n dimensions, and each dimension has exactly p distinct values in the base cuboid. Assume that there are no concept hierarchies associated with the dimensions.

    (a) What is the maximum number of cells possible in the base cuboid?

    (b) What is the minimum number of cells possible in the base cuboid?

    (c) What is the maximum number of cells possible (including both base cells and aggregate cells) in the C data cube?

    (d) What is the minimum number of cells possible in C?
20. Suppose that a data warehouse consists of the four dimensions, **date, spectator, location, and game,** and the two measures, **count and charge**, where charge is the fare that a spectator pays when watching a game on a given date. Spectators may be students, adults, or seniors, with each category having its own charge rate.

    (a) Draw a star schema diagram for the data warehouse.

    (b) Starting with the base cuboid [date, spectator, location, game], what specific OLAP operations should one perform in order to list the total charge paid by student spectators at GM Place in 2010?